Georgy Dunaev

June 24, 2024

0.1 "Paradoxes" of class theory

Theorem 0.1. In classical and even in intuitionistic logic, the following holds:

$$\neg(\exists W \forall Q : Q \in W \iff Q \notin Q)$$

Proof.
By contradiction, assume $(\exists W \forall Q : Q \in W \iff Q \notin Q)$. To show: $\bot$.
Take such W.
Set Q:=W.
Therefore $W \in W \iff W \notin W$. (we already could say obtained a contradiction, but want to show that it doesn't need LEM.)
From **??** where $P := (W \in W)$, we have: $\neg\neg((W \in W) \vee (W \notin W))$.
Therefore the goal is: $W \in W \iff W \notin W \vdash \neg((W \in W) \vee (W \notin W))$
Now, using backward reasoning:
$W \in W \iff W \notin W, (W \in W) \vee (W \notin W) \vdash \bot$.
Subgoal1:
$W \in W \iff W \notin W, (W \in W) \vdash \bot$.
$W \in W \iff W \notin W, (W \in W) \vdash (W \notin W)$.
$W \in W \iff W \notin W, (W \in W) \vdash (W \in W)$ (by assumption).
Subgoal2:
$W \in W \iff W \notin W, (W \notin W) \vdash \bot$.
$W \in W \iff W \notin W, (W \notin W) \vdash (W \in W)$.
$W \in W \iff W \notin W, (W \notin W) \vdash (W \notin W)$ (by assumption).
Done.

Theorem 0.2. In classical and even in intuitionistic logic, the following holds:

$$\forall Y : Y \notin Y$$

Proof.
Let $C := \{W | W \notin W\}$.
Goal $\mathcal{P}C \subseteq C$.
Fix class B. Assume $B \in \mathcal{P}C$.
$(B \in \mathcal{P}C) \Longrightarrow (B \text{ set}) \wedge (B \subseteq C)$.
Claim: $\forall b \in B : b \notin b$.
$(b \in B) \ \ \& \ \ (B \subseteq C) \Longrightarrow (b \in C)$.
$(b \in C) \Longrightarrow (b \notin b)$.
Claim: $B \notin B$.
Assume opposite $B \in B$ and derive a contradiction.
$(B \in B) \Longrightarrow (B \notin B)$ (by previous claim).

$(B \in B)$ & $(B \notin B) \implies \bot$.
Contradiction obtained, so the claim $B \notin B$ is proven.
$(B \text{ set})$ & $(B \notin B) \implies (B \in C)$.
We are done proving the goal $\mathcal{P}C \subseteq C$.
$\mathcal{P}C \subseteq C \implies V \subseteq C$ (by epsilon induction).
Goal: $\forall Y : Y \notin Y$.
Fix Y.
Case 1: $(Y \text{ set})$.
$(Y \text{ set}) \implies (Y \in V)$.
$(Y \in V) \wedge (V \subseteq C) \implies (Y \in C)$.
$(Y \in C) \implies (Y \notin Y)$.
Case 2: $\neg(Y \text{ set})$.
$\neg(Y \text{ set}) \implies \neg(\exists Q : Y \in Q)$.
$\neg(\exists Q : Y \in Q) \implies (\forall Q : Y \notin Q)$.
$(\forall Q : Y \notin Q) \implies (Y \notin Y)$. (by $Q := Y$)
Done.

Theorem 0.3. If intersection of a class A is a set, then A is nonempty.

$$\left(\left(\bigcap A\right) \text{ set}\right) \longrightarrow (\exists w : w \in A)$$

Proof.
New goal:

$$\neg\exists w : w \in A \longrightarrow \neg(\left(\bigcap A\right) \text{ set})$$

Proof of the claim:

$$\neg\exists w : w \in A \implies (A = \emptyset)$$

$$(A = \emptyset) \implies \left(\bigcap A\right) = \left(\bigcap \emptyset\right)$$

$$\left(\bigcap \emptyset\right) = V$$

$$\left(\bigcap A\right) = \left(\bigcap \emptyset\right) \quad \& \quad \left(\bigcap \emptyset\right) = V \implies \left(\bigcap A\right) = V$$

$$(V \notin V) \quad (\text{by thm } \mathbf{??} \text{ with } Y := V)$$

$$(V \notin V) \implies \neg(V \text{ set}) \quad (\text{by thm } \mathbf{??})$$

$$\left(\bigcap A\right) = V \quad \& \quad \neg(V \text{ set}) \implies \neg(\left(\bigcap A\right) \text{ set})$$

The claim is proven.

$$\left(\left(\bigcap A\right) \text{ set}\right) \implies (\exists w : w \in A) \quad (\text{from claim by contraposition})$$

Done.

Theorem 0.4. In classical and even in intuitionistic logic, the following holds:

$$\forall(Y \text{ set}) : \neg(\mathcal{P}Y \subseteq Y)$$

Proof.

$$C := \{w | \neg(\mathcal{P}w \subseteq w)\}$$

Claim $\mathcal{P}C \subseteq C$:
Fix $B \in \mathcal{P}C$. To show: $B \in C$
$(B \in \mathcal{P}C) \Longrightarrow (B \text{ set}) \wedge (B \subseteq C)$.
Assume opposite: $B \notin C$, and derive a contradiction.
$(B \notin C) \Longrightarrow (\neg(B \text{ set})) \vee (\neg\neg(\mathcal{P}w \subseteq w))$.
$(B \text{ set}), (\neg(B \text{ set})) \vee (\neg\neg(\mathcal{P}B \subseteq B)) \Longrightarrow \neg\neg(\mathcal{P}B \subseteq B)$.
$\neg\neg(\mathcal{P}B \subseteq B) \Longrightarrow (\mathcal{P}B \subseteq B)$.
$(\mathcal{P}B \subseteq B) \wedge (B \subseteq C) \Longrightarrow (\mathcal{P}B \subseteq C)$.
$(B \in \mathcal{P}B)$.
$(B \in \mathcal{P}B) \wedge (\mathcal{P}B \subseteq C) \Longrightarrow (B \in C)$.
$(B \in C) \wedge (B \notin C) \Longrightarrow \bot$
Contradiction obtained, so the claim $\mathcal{P}C \subseteq C$ is proven.
$(\mathcal{P}C \subseteq C) \Longrightarrow (V \subseteq C)$ (by regularity axiom).
Fix $(Y \text{ set})$.
$(Y \text{ set}) \Longrightarrow (Y \in V)$.
$(Y \in V) \wedge (V \subseteq C) \Longrightarrow (Y \in C)$.
$(Y \in C) \Longrightarrow \neg(\mathcal{P}Y \subseteq Y)$.
Done.

Theorem 0.5. A conglomerate of all classes is not a class. It "does not exist as class".

$$\neg(\exists W \forall Q : Q \in W)$$

$$(\forall W \exists Q : Q \notin W)$$

Proof.
Fix W.
Set $Q := W$.
By **??**, $W \notin W$.
So $Q \notin W$.
Done.